Width-independent learning bounds for ReLU, GELU, and other Lipschitz activations

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Abstract

We prove fixed-depth polynomial sample-complexity bounds for Bayesian neural networks with any fixed globally Lipschitz activation that vanishes at zero. The class includes ReLU, exact GELU, SiLU, and tanh. A bounded-weight teacher of width C can be learned at every learner width $N \geq C$, with one polynomial sample bound simultaneously valid for every Gaussian prior scale at least as strong as mean field. In particular, the same bound covers all exponents $\gamma \geq 1$, including the deep mean-field endpoint, without deterioration as γ approaches one. The prior is conditioned before training on a strict absolute output margin at every Boolean input; training conditions on exact agreement with noiseless labels. A width-independent capacity estimate handles unbounded, nonhomogeneous activations by allowing auxiliary neuron-specific Lipschitz functions. Spectral truncation controls stronger priors. At mean field, a Gaussian projection separates a bounded bulk from a tail of controlled reproducing-kernel energy, and stable covariance reconstruction yields a finite auxiliary implementation. The actual Bayesian learner continues to use its single fixed activation. We also give a simpler strong-prior proof and an exact same-depth GELU implementation of layered Boolean circuits with at most twice the width. The quantitative polynomials are loose and retain poor depth dependence.

1 Introduction

Can a neural-network posterior learn a small implementing network without paying for the much larger width of the learner? Bounded-activation proofs often compress hidden states by sampling neurons. ReLU and GELU require different care: individual hidden states can be arbitrarily large, and GELU does not permit the exact rescaling identities available for ReLU.

We prove that neither boundedness nor homogeneity is necessary for a fixed-depth result. A globally Lipschitz activation ϕ satisfying $\phi(0) = 0$ is enough. The proof controls the product of layer Frobenius norms, embeds the resulting networks into an auxiliary class with neuron-specific Lipschitz activations, and bounds that larger class using a one-dimensional activation cover. This gives a finite class of globally margin-separated Boolean functions at each fixed energy, simultaneously across all widths.

That capacity statement has two applications. First, in the strong-prior limit, Gaussian rates suppress functions whose minimum energy exceeds the target's. Second, at every mean-field-or-stronger scale, a Gaussian functional compression argument replaces typical margin-separated draws by auxiliary networks of controlled complexity. For small scales, spectral truncation discards the operator-small bulk. At larger scales, including mean field, bounded exponential tilts describe the

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regular part of the hidden covariance, while a separate reproducing-kernel argument retains rare large neurons. Combining the two constructions gives one polynomial bound across the entire range of scales.

These are information-theoretic guarantees for exact Bayesian conditioning, not claims about efficient posterior computation. They concern iid noiseless learning on the Boolean cube and the specified global-margin conditioning. The polynomial degree depends on depth; a polynomial-depth theorem requires additional ideas or restrictions. Established tools used below include Rademacher contraction and peeling, Gaussian concentration, and finite-class Occam learning [2, 1, 3]. We give the particular estimates and normalization needed here.

2 Model and the results

Fix input dimension $d \geq 1$, depth $L \geq 2$, and $H = L - 1$ hidden layers. Let $\mathcal{X} = \{-1, 1\}^d$, $\bar{x} = (1, x_1, \ldots, x_d)$, and fix an activation $\phi : \mathbb{R} \to \mathbb{R}$ with $\phi(0) = 0$ and Lipschitz constant at most $\lambda \geq 1$. A width-N network is

$$h^0(x) = \bar{x}, \qquad h^s(x) = \phi(W^s h^{s-1}(x)) \quad (1 \leq s < L), \qquad z_w(x) = v^\top h^{L-1}(x),$$

where activation is coordinatewise and every hidden layer has width N. There are no separate internal biases; constant wires can carry offsets. Define the weighted Gaussian energy

$$\kappa_N(w) = \frac{d}{N} \left\| W^1 \right\|_F^2 + \sum_{s=2}^{L-1} \|W^s\|_F^2 + N \|v\|_2^2 . \tag{1}$$

The raw prior $\mu_{N,\sigma}$ has independent Gaussian coordinates with variances

$$(N/d)\sigma^2 \quad \text{in the first layer,} \qquad \sigma^2 \quad \text{internally,} \qquad \sigma^2/N \quad \text{in the readout.}$$

Its density is proportional to $\exp[-\kappa_N(w)/(2\sigma^2)]$. Prescribed exponent γ means $\sigma^2 = N^{-\gamma}$.

For a Boolean function g, set

$$U_{N,g} = \{w : g(x) z_w(x) > 1/2 \text{ for every } x \in \mathcal{X}\}, \qquad U_N = \bigcup_g U_{N,g}.$$

Condition the raw prior on U_N before seeing data. This induces the function prior

$$\pi_{N,\sigma}(g) = \frac{\mu_{N,\sigma}(U_{N,g})}{\mu_{N,\sigma}(U_N)}.$$

Given iid examples $S_n = ((X_i, f(X_i)))_{i=1}^n$ with $X_i \sim D$, condition again on agreement with all labels, obtaining $\Pi_{N,\sigma}(\cdot \mid S_n)$. Its average sign error is

$$R_D(\Pi) = \mathbb{E}_{g \sim \Pi} \mathbb{P}_{X \sim D}\{g(X) \neq f(X)\}.$$

The distribution D may be any distribution on the cube.

Assume f has a width-C, depth-L teacher with all individual weights bounded in absolute value by C, satisfying $f(x) z_*(x) > 1/2$ for every input. This is the only teacher assumption. For an arbitrary activation it need not hold for every Boolean circuit; we verify it with a small implementation for ReLU and exact GELU in Section 9.

Theorem 2.1 (Summary of learning guarantees)**.** *For fixed L and λ, there is a sample bound polynomial in $d, C, \varepsilon^{-1}, \log(1/\delta)$, independent of learner width, that gives $R_D(\Pi_{N,\sigma}(\cdot \mid S_n)) \le \varepsilon$ with probability at least $1-\delta$, simultaneously for every $N \ge C$ and $0 < \sigma^2 \le 1/N$. Thus one polynomial works for every prescribed exponent $\gamma \ge 1$ in $\sigma^2 = N^{-\gamma}$, and also covers the strong-prior limit at every fixed width.*

Theorem 8.2 gives the quantitative bound through constructive polynomial budgets. Theorem 4.3 gives a simpler strong-prior argument with a smaller explicit bound.

The probability statements are over the training inputs. The posterior may mix tied implementations or tied minimum-energy functions. No uniqueness is assumed. If $p_S(x)$ is the posterior probability of the correct label, then

$$D\{x : p_S(x) \le q\} \le \frac{R_D(\Pi)}{1-q} \qquad (0 < q < 1). \tag{2}$$

Thus using $\varepsilon/3$ in an average-risk bound gives correct-label probability strictly greater than $2/3$ except on an input set of measure at most ε. Posterior-majority error is at most twice the average sign error.

Examples include ReLU and tanh with $\lambda = 1$; exact GELU $\phi(t) = t\Phi(t)$ and SiLU $\phi(t) = t/(1+e^{-t})$ with $\lambda = 2$; and centered softplus $\phi(t) = \log(1+e^t) - \log 2$ with $\lambda = 1$. A nonzero activation offset is not covered by the theorem as stated.

3 A capacity estimate without boundedness or homogeneity

Let Ψ be the family of all globally 1-Lipschitz real functions ψ with $\psi(0) = 0$.

Lemma 3.1 (A scalar activation cover)**.** *For $R > 0$ and $0 < \eta \le R$, the restrictions of Ψ to $[-R, R]$ have a supremum-norm η-net with M_η centers that themselves belong to Ψ, satisfying*

$$\log(2M_\eta) \le 14R/\eta.$$

Proof. Put $J = \lceil 3R/\eta \rceil$ and $\Delta = R/J$. Round the values of ψ at the grid points $j\Delta$, $-J \le j \le J$, to the nearest multiple of Δ. The value at zero has code zero, and successive rounded integers differ by at most two. There are at most 5^{2J} codes. From each nonempty code choose one actual member of Ψ as its representative. Functions with the same code differ by at most Δ at grid points and at most $3\Delta \le \eta$ everywhere on the interval. Finally

$$\log(2M_\eta) \le \log 2 + 2(3R/\eta + 1)\log 5 \le 14R/\eta.$$

Choosing actual representatives preserves the Lipschitz constant exactly. □

Lemma 3.2 (Convex contraction)**.** *For a bounded set $T \subset \mathbb{R}^m$, a fixed $\psi \in \Psi$, and $a > 0$,*

$$\mathbb{E}_\epsilon \exp\left(a \sup_{t\in T} \sum_i \epsilon_i \psi(t_i)\right) \le \mathbb{E}_\epsilon \exp\left(a \sup_{t\in T} \sum_i \epsilon_i t_i\right).$$

The same holds with any convex nondecreasing function in place of the exponential.

Proof. Replace one Rademacher coordinate at a time. With the other coordinates fixed, write a_t for their contribution and

$$X_\pm = \sup_t (a_t \pm \psi(t_k)), \qquad Y_\pm = \sup_t (a_t \pm t_k).$$

Because $|\psi(t_k)| \le |t_k|$, one has $\max(X_+, X_-) \le \max(Y_+, Y_-)$. Also

$$X_+ + X_- \le \sup_{t,u}(a_t + a_u + |t_k - u_k|) = Y_+ + Y_-.$$

For two numbers, these maximum and sum inequalities imply the corresponding inequality for the sum of any convex nondecreasing function. Average over the coordinate and iterate. Suprema follow by finite approximation. □

Let $\mathcal{G}_s$ consist of all vector-valued feature maps of depth s whose layer matrices have Frobenius norm at most one and whose neurons use arbitrary, independently chosen members of Ψ. Widths can be any finite integers. Put $\mathcal{G}_0 = \{x \mapsto x\}$. All feature vectors have norm at most R on inputs of norm at most R.

Lemma 3.3 (Width-independent feature complexity)**.** *For $H \ge 1$, inputs $x_1, \dots, x_m$ of Euclidean norm at most R, and independent uniform signs ϵ_i,*

$$\frac{1}{m}\mathbb{E}_\epsilon \sup_{h \in \mathcal{G}_H} \left\| \sum_i \epsilon_i h(x_i) \right\|_2 \le 8HRm^{-1/3}.$$

Proof. Write $Z_s = \sup_{h \in \mathcal{G}_s} \| \sum_i \epsilon_i h(x_i)\|_2$. Decompose each row of the final matrix as $r_j u_j$, where $r_j \ge 0$, $\|u_j\|_2 \le 1$, and $\sum_j r_j^2 \le 1$. For $r_j > 0$, the map $t \mapsto \psi_j(r_j t)/r_j$ is again in Ψ; zero rows contribute zero. Consequently

$$Z_s \le \sup_{h \in \mathcal{G}_{s-1},\, \|u\|_2 \le 1,\, \psi \in \Psi} \left| \sum_i \epsilon_i \psi(u \cdot h(x_i)) \right|.$$

Replace ψ by a center from Lemma 3.1, at an additive cost $m\eta$. For each center, handle the absolute value by its two signs and apply Lemma 3.2. Since $\sup_{h,u} \sum_i \epsilon_i u \cdot h(x_i) = Z_{s-1}$, taking a finite union gives

$$\mathbb{E}e^{aZ_s} \le 2M_\eta e^{am\eta}\mathbb{E}e^{aZ_{s-1}} \qquad (a > 0).$$

At depth zero, $\mathbb{E}Z_0 \le R\sqrt{m}$. Changing one sign changes Z_0 by at most $2R$, so the bounded-differences exponential estimate gives

$$\mathbb{E}e^{aZ_0} \le \exp(aR\sqrt{m} + a^2 mR^2/2).$$

Iterate H times and optimize a to obtain

$$\mathbb{E}Z_H \le Hm\eta + R\sqrt{m} + R\sqrt{2mH\log(2M_\eta)}.$$

Taking $\eta = Rm^{-1/3}$ and using $1 + H + \sqrt{28H} \le 8H$ proves the claim. No bound here depends on the number of neurons. All suprema are well-defined on the finite Rademacher sample space; equivalently one can take finite subclasses and pass to monotone limits. □

Proposition 3.4 (Product-norm bound for Lipschitz activations)**.** *Consider networks with $H = L-1$ hidden layers, with a possibly different zero-at-zero λ-Lipschitz activation at each neuron, and with product of layer Frobenius norms at most P. Their empirical Rademacher complexity is at most*

$$8H\lambda^H PRm^{-1/3}$$

on any m inputs of norm at most R, independently of the finite widths.

Proof. Write $m_s = \|W^s\|_F$, treating the readout as layer L, and let $\phi_{s,j}$ be the activation of neuron j in hidden layer s. If some $m_s = 0$, the network is zero. Otherwise put $q_0 = 1$, $q_s = \lambda^s \prod_{j \le s} m_j$, and $g_s = h_s/q_s$. With $A_s = W^s/m_s$ one has the exact coordinate identities

$$g_{s,j} = \psi_{s,j}((A_s g_{s-1})_j), \qquad \psi_{s,j}(t) = \frac{\phi_{s,j}(m_s q_{s-1} t)}{\lambda m_s q_{s-1}} \in \Psi.$$

The output is $\lambda^H \prod_s m_s$ times a norm-one linear readout of g_H. Apply Lemma 3.3. □

Remark 3.5. This normalization embeds a fixed-activation network into a larger auxiliary class. It changes the scalar activations. It is not a parameter rescaling inside the original GELU learner and does not assume GELU is homogeneous. The actual prior and teacher continue to use the single fixed activation ϕ.

For $K > 0$ let $\mathcal{F}(K)$ contain every Boolean function admitting a depth-L, globally strict-margin-$1/2$ implementation of energy at most K, at some finite equal hidden width, allowing each hidden neuron its own activation vanishing at zero with Lipschitz constant at most λ. This is an auxiliary counting class, larger than the fixed-ϕ learner's function class. Proposition 3.4 and its proof apply unchanged to this neuron-specific class. Define

$$v(K) = \left\lceil \left[16 H \lambda^H \sqrt{\frac{d+1}{d}} \left(\frac{K}{L}\right)^{L/2} \right]^3 \right\rceil, \qquad \mathcal{B}(K) = v(K) \log(2^d + 1). \tag{3}$$

Corollary 3.6 (Counting globally margin-separated functions)**.** *The common class across all widths satisfies* $\log |\mathcal{F}(K)| \le \mathcal{B}(K)$.

Proof. The squared norms of $\sqrt{d/N} W^1, W^2, \ldots, W^{L-1}, \sqrt{N} v^\top$ sum to κ_N and their product equals $d \prod_s \|W^s\|_F^2$. Arithmetic–geometric mean yields

$$\prod_s \|W^s\|_F \le d^{-1/2} (K/L)^{L/2}.$$

The appended inputs have norm $\sqrt{d+1}$. Shattering m cube inputs with global margin $> 1/2$ forces the empirical Rademacher complexity of the underlying real-output class to be at least $1/2$. Proposition 3.4 bounds its VC dimension by $v(K)$. The Sauer bound on the 2^d-point domain gives $|\mathcal{F}(K)| \le (2^d + 1)^{v(K)}$. □

4 The strong-prior learning theorem

Lemma 4.1 (Exact width enlargement)**.** *The target teacher can be implemented at every width* $N \ge C$ *with the same real-valued output and energy at most*

$$B_* = d(d+1)C^2 + LC^4.$$

Proof. Let $k = \lfloor N/C \rfloor$. Make k copies of each hidden neuron, copy the first-layer incoming weights, divide all internal weights by k between each pair of copied populations, and divide the readout weights by k. Fill remaining slots with zero neurons; $\phi(0) = 0$ makes padding harmless. Each copy computes exactly the original neuron's activation. The first-layer energy is at most $d(d+1)C^2$, internal Frobenius energies are unchanged and at most C^4 each, and readout energy is at most $2C^4$ because $N/k < 2C$. This gives the claimed sum. □

Lemma 4.2 (Finite-dimensional Gaussian rate)**.** *For fixed N and nonempty $U_{N,g}$, put $k_N(g) = \inf_{w \in U_{N,g}} \kappa_N(w)$. Then*

$$\lim_{\sigma \downarrow 0} 2\sigma^2 \log \mu_{N,\sigma}(U_{N,g}) = -k_N(g).$$

Proof. In weighted coordinates the raw law is an isotropic Gaussian. A small ball around any point of the open event gives the lower bound with rate equal to that point's squared norm, and then to the infimum. The closed-set Gaussian upper bound applies to its closure. A point of this closure has signed outputs at least $1/2$ everywhere; multiplying its readout by any factor greater than one places it in $U_{N,g}$ with arbitrarily small increase in energy. Hence the closure and the open event have equal energy infima. The elementary Gaussian upper bound can equivalently be obtained from the chi-square tail outside the ball with squared radius $k_N(g)$. □

Theorem 4.3 (Strong-prior learning)**.** *Let B_* be as in Lemma 4.1. For every $N \geq C$ and $0 < \varepsilon < 1$, there is $\sigma_0(N) > 0$ such that for $0 < \sigma < \sigma_0(N)$,*

$$n \geq \frac{2}{\varepsilon} \left[\mathcal{B}(B_* + 1) + \log(1/\delta) \right]$$

implies $R_D(\Pi_{N,\sigma}(\cdot \mid S_n)) \leq \varepsilon$ with sample probability at least $1 - \delta$, for every D and $0 < \delta < 1$. The threshold can depend on d, L, λ, N, the teacher, and ε, but not on D, n, δ. A common sample event works at every $N \geq C$, using the width-specific variance thresholds.

Proof. At fixed width, retain functions satisfying $k_N(g) \leq k_N(f)$. They are all in $\mathcal{F}(B_* + 1)$ by Lemma 4.1 and the definition of infimum. There are finitely many Boolean functions, so Lemma 4.2 makes the total raw mass of the remaining realizable functions divided by the target mass tend to zero. Choose $\sigma_0(N)$ so the ratio is at most $\varepsilon/2$. Global-margin normalization cancels. On every noiseless training history, the posterior denominator includes the target; thus posterior mass outside the retained class remains at most $\varepsilon/2$.

By a finite-class union bound, with probability at least $1 - \delta$, every member of $\mathcal{F}(B_* + 1)$ that agrees with all n labels has true error at most $\varepsilon/2$. The stated sample bound suffices since a function of error greater than $\varepsilon/2$ survives with probability at most $e^{-n\varepsilon/2}$. On this common event, posterior average error is at most ε. This event does not depend on width. □

At fixed L, λ, this bound is polynomial in the indicated variables; its crude power of C is at most $6L$. Positive homogeneity allows a sharper estimate for ReLU, but is unnecessary for the existence of a polynomial. The strong-prior limit is taken at fixed width before invoking the conclusion, and does not assert that one variance threshold works at all widths.

5 Target mass and spectral compression

This section extends the Gaussian spectral argument to unbounded activations. Let $\phi(0) = 0$ and let ϕ be λ-Lipschitz; replace λ by $\max\{1, \lambda\}$, so throughout $\lambda \geq 1$. For $\gamma \geq 1$, the independent coordinate variances are $N^{1-\gamma}/d$ in the first layer, $N^{-\gamma}$ internally, and $N^{-1-\gamma}$ at the readout. Write $U_{N,g}$ for the globally strict-margin event $g(x) z_w(x) > 1/2$ for every cube input. The normalized energy is

$$\kappa_N(w) = \frac{d}{N} \|W^1\|_F^2 + \sum_{s=2}^{L-1} \|W^s\|_F^2 + N \|v\|_2^2.$$

No uniform bound on ϕ or on individual hidden activations is imposed.

5.1 A target cylinder for linearly growing activations

Let a depth-L, width-C teacher have every weight bounded by C and strict output margin $1/2$. Multiplying only its readout by three gives strict margin $3/2$. Set

$$\begin{aligned} M &= (d+1)(\lambda C^2)^{L-1}, & b &= 2+\sqrt{2\log(2L)}, \\ B &= 2+\lambda(\sqrt{2}C^2+\sqrt{C}+b), & E &= \lambda M(1+LC)B^{L-2}, \\ \rho &= \Big[4\{(3\sqrt{2}C^2+1)E+M\}\Big]^{-1}, & & \\ B_0 &= d(d+1)C^2+2(L-2)C^4+36C^4, & D_0 &= d+2+(L-2)C, \\ A &= 2B_0+4D_0/\rho+2. & & \end{aligned}$$

These quantities are independent of learner width and polynomially bounded in d, C for fixed L, λ. In particular $0<\rho\le 1$ and $A\ge 1$.

Proposition 5.1 (Unbounded-activation target mass)**.** *At every $N\ge C$ and every $\gamma\ge 1$, the raw Gaussian probability of the target event satisfies*

$$\mu_{N,\gamma}(U_{N,f}) \ge \exp(-AN^{\gamma}/2).$$

Proof. Partition the N neurons into C groups of sizes k_a differing by at most one. Then $N/k_a\le 2C$ and $k_a/k_b\le 2$. Clone the teacher by copying each first-layer row throughout its group, replacing an internal weight B^s_{ba} by B^s_{ba}/k_a on edges from group a to group b, and replacing readout weight v_b by v_b/k_b. This preserves the exact real-valued function for any activation. Every cloned hidden state is group-constant. Let P project onto that subspace.

Because $|\phi(u)|\le\lambda|u|$, all teacher hidden coordinates are bounded in magnitude by M: the first layer is bounded by $\lambda C(d+1)$, and each next hidden layer multiplies this bound by at most λC^2. The cloned internal operator norms are at most $\sqrt{2}C^2$, and the scaled readout norm is at most $\sqrt{N}\|v_*\|_2\le 3\sqrt{2}C^2$.

Put $t=N^{(1-\gamma)/2}\le 1$. In the first layer constrain each $\sqrt{d}W^1_{ij}$ within ρt of its cloned value. In each internal row, constrain its C coordinates along the normalized group indicators, after multiplying the row by $\sqrt{N}$, within ρt of their cloned values. Finally constrain each Nv_j within ρt of its cloned value. All these are independent scalar Gaussian coordinates of variance t^2. Write the resulting internal matrices as

$$W^s = W^s_* + R^s + E^s, \qquad R^s = R^sP, \quad E^sP=0.$$

The constrained error obeys $\|R^s\|_F\le\rho t\sqrt{C}$; the free part is $(t/\sqrt{N})G^s(I-P)$, independently of the constrained coordinates, with standard Gaussian G^s. The Gaussian operator-norm bound and a union bound give conditional probability at least $1/2$ that $\|E^s\|_{\mathrm{op}}\le bt$ for all internal layers.

On this event put $e_s(x)=N^{-1/2}\|h^s_w(x)-h^s_*(x)\|_2$. Uniformly over the cube,

$$e_1\le\lambda\rho(d+1), \qquad e_s\le Be_{s-1}+\lambda\rho\sqrt{C}\,M.$$

The second estimate uses $E^sh^{s-1}_*=0$, the bound on the teacher hidden states, and Lipschitz continuity. It does not bound the perturbed states by one. Since $M\ge d+1$ and $B\ge 1$, induction gives $e_{L-1}\le\rho E$. The readout error has scaled norm at most ρ, so

$$\sup_x|z_w(x)-z_*(x)|\le(3\sqrt{2}C^2+1)e_{L-1}+\rho M\le 1/4.$$

Thus every such draw lies in $U_{N,f}$ with room to spare.

There are ND_0 constrained coordinates and their squared centers sum to at most NB_0. For $Z \sim N(0, t^2)$,

$$\mathbb{P}\{|Z - a| \le \rho t\} \ge \frac{2\rho}{\sqrt{2\pi}} e^{-a^2/t^2 - \rho^2}.$$

Multiplying these bounds, retaining the independent operator-norm event, and using $N \le N^\gamma$ gives

$$-\log \mu_{N,\gamma}(U_{N,f}) \le N^\gamma B_0 + ND_0 \log \frac{e\sqrt{2\pi}}{2\rho} + \log 2 \le AN^\gamma/2.$$

Here $\log(e\sqrt{2\pi}/(2\rho)) \le 2/\rho$ for $0 < \rho \le 1$. □

5.2 Spectral truncation with controlled hidden RMS norms

Lemma 5.2 (Gaussian spectral tail)**.** *If W is an $N \times N$ matrix with independent $N(0, \sigma^2)$ entries, $a \ge 8\sigma\sqrt{N}$, and $W_{>a}$ retains singular values greater than a, then*

$$\mathbb{P}\{\|W_{>a}\|_F^2 > K\} \le e^{-K/(8\sigma^2)}.$$

Proof. The Gaussian operator-norm inequality $\mathbb{P}\{\|W\|_{\mathrm{op}} > 2\sigma\sqrt{N} + \sigma u\} \le e^{-u^2/2}$ shows that the operator-norm ball of radius $a/2$ has probability greater than $1/2$. Its Frobenius distance from W is $D(W) = [\sum_j (s_j(W) - a/2)_+^2]^{1/2}$. Gaussian isoperimetry gives $\mathbb{P}\{D(W) > r\} \le e^{-r^2/(2\sigma^2)}$. Since $s \le 2(s - a/2)$ when $s > a$, one has $\|W_{>a}\|_F \le 2D(W)$. Take $r = \sqrt{K}/2$. □

For $K \ge 1$, define

$$a_K = \left[16L\lambda^L(K+1)(\sqrt{K}+1)^{L-2}\right]^{-1}, \qquad J_K = (L+3)K.$$

Use the auxiliary class $\mathcal{F}(J)$ from Corollary 3.6; it includes the Boolean functions admitting a strict margin-$1/2$, depth-L, fixed-ϕ implementation of energy at most J at some finite width.

Proposition 5.3 (Aggregate functional tail)**.** *Suppose $\gamma > 1$ and*

$$N^{\gamma-1} \ge 8(d+1), \qquad N^{\gamma-1} \ge 64 a_K^{-2}.$$

Then

$$\mu_{N,\gamma}\left(\bigcup_{g \notin \mathcal{F}(J_K)} U_{N,g}\right) \le Le^{-KN^\gamma/8}.$$

Proof. Outside an event of probability at most $Le^{-KN^\gamma/8}$,

$$\frac{d}{N}\|W^1\|_F^2 \le K, \quad N\|v\|_2^2 \le K, \quad \|(W^s)_{>a_K}\|_F^2 \le K \quad (2 \le s < L).$$

Indeed, the first and last random quantities are respectively $N^{-\gamma}\chi^2_{N(d+1)}$ and $N^{-\gamma}\chi^2_N$. For $u \ge 8D$, Markov's inequality with exponential parameter $3/8$ gives $\mathbb{P}\{\chi^2_D > u\} \le 2^D e^{-3u/8} \le e^{-u/4}$; the first width condition therefore suffices. The second width condition permits Lemma 5.2 on each internal matrix.

Replace each internal matrix by its $> a_K$ part. Retained operator norms are at most $\sqrt{K}$ and discarded operator norms at most $a_K \le 1$. For the original hidden states write $M_s = N^{-1/2} \sup_x \|h_w^s(x)\|_2$. The zero-at-zero and Lipschitz assumptions imply

$$M_s \le \sqrt{2K}\, \lambda^s (\sqrt{K}+1)^{s-1}.$$

In particular individual neurons need not be bounded, but their normalized Euclidean norm is controlled uniformly over inputs. If e_s is the normalized Euclidean difference between original and truncated hidden states, then

$$e_1 = 0, \qquad e_s \le \lambda\sqrt{K}\, e_{s-1} + \lambda a_K M_{s-1}.$$

Induction gives, for $s \ge 2$,

$$e_s \le a_K(s-1)\sqrt{2K}\lambda^s(\sqrt{K}+1)^{s-2}.$$

The output discrepancy is consequently at most

$$a_K(L-2)\sqrt{2}K\lambda^{L-1}(\sqrt{K}+1)^{L-3} < 1/8$$

when $L \ge 3$. It is zero when $L = 2$. A globally margin-separated original draw therefore retains strict margin $3/8$ after truncation. Doubling the readout gives strict margin $3/4$ and energy at most $K + (L-2)K + 4K = J_K$. Its Boolean function belongs to $\mathcal{F}(J_K)$, proving the claimed aggregate tail. □

The small spectral cutoff requires a sufficiently strong scale. At mean field, the Gaussian bulk has operator norm of constant order; the next sections control its functional effect through covariance reconstruction.

6 Gaussian bulk, tail atoms, and stable exponential tilts

Let Q be a positive semidefinite matrix on a finite input set of size m, with $\max_x Q_{xx} \le M$ and $M \ge 1$. Its reproducing-kernel Hilbert space $\mathcal{H}_Q$ is its range, with norm $\|u\|_Q^2 = u^\top Q^+ u$. Thus $|u_x| \le \sqrt{M}\|u\|_Q$. All input-set dependence in the quantitative bounds below is through $\log(2m)$.

Lemma 6.1 (Gaussian projection onto a bounded bulk)**.** *Let $Z_1, \dots, Z_N$ be independent $N(0, Q)$ vectors. Put $B = \sqrt{2M\log(4m)}$ and $A = 4(K + \log 2)$, where $K > 0$. There are measurable decompositions $Z_i = a_i + u_i$ with $\|a_i\|_\infty \le B$ and $u_i \in \mathcal{H}_Q$ such that*

$$\mathbb{P}\left\{N^{-1}\sum_i \|u_i\|_Q^2 > A\right\} \le e^{-NK}.$$

Proof. The closed convex set $\{a \in \mathcal{H}_Q : \|a\|_\infty \le B\}$ has Gaussian probability at least $1/2$ by a union bound over the m coordinates. Let $a(z)$ be its metric projection in $\mathcal{H}_Q$, and let $u(z) = z - a(z)$. Gaussian isoperimetry on the support of Q gives $\mathbb{P}\{\|u(Z)\|_Q > s\} \le e^{-s^2/2}$. Thus $\mathbb{E}e^{\|u(Z)\|_Q^2/4} \le 2$ by integration of the tail. Independence and exponential Markov give $\exp(N\log 2 - NA/4) = e^{-NK}$. The projection is continuous and hence measurable. For $Q = 0$ take $a = u = 0$. □

Lemma 6.2 (General Lipschitz version for auxiliary activations)**.** *Suppose instead only that $\phi(0) = 0$ and ϕ is globally λ-Lipschitz. Keep B, A and the Gaussian projections $Z_i = a_i + u_i$ from Lemma 6.1, and take $T \ge 2B$. Let $r : \mathbb{R}^m \to [0,1]$ equal one on $\|z\|_\infty \le T$, and define $Q_{\text{bulk}} = N^{-1}\sum_i r(Z_i)^2\phi(Z_i)\phi(Z_i)^\top$. Then, on the projection-good event, the empirical activation covariance is within $6\lambda^2 BMA/T$ of its same bounded bulk plus*

$$R = \frac{1}{N}\sum_i (1 - r(Z_i)^2)\phi(u_i)\phi(u_i)^\top.$$

Writing $\delta_i = \|u_i\|_{\mathcal{H}_Q}$ and omitting zero terms, this is

$$R = \frac{1}{N}\sum_i (1 - r(Z_i)^2)\delta_i^2 \psi_i(v_i)\psi_i(v_i)^\top, \qquad v_i = u_i/\delta_i, \quad \psi_i(t) = \phi(\delta_i t)/\delta_i.$$

Every v_i has unit RKHS norm, every ψ_i is λ-Lipschitz and zero at zero, and the nonnegative mixture weights sum to at most A. The same good event bounds the next covariance diagonal by $2\lambda^2 B^2 + 2\lambda^2 MA$.

Proof. Lipschitz continuity gives $\|\phi(Z_i) - \phi(u_i)\|_\infty \le \lambda B$ and $\|\phi(u_i)\|_\infty \le \lambda\sqrt{M}\delta_i$. If $w_i = 1 - r(Z_i)^2 > 0$, then $\sqrt{M}\delta_i \ge T - B \ge T/2$. Hence

$$N^{-1}\sum_i w_i \le 4MA/T^2, \qquad N^{-1}\sum_i w_i\delta_i \le 2\sqrt{M}A/T.$$

The outer-product difference is therefore at most $4\lambda^2 BMA/T + 4\lambda^2 B^2 MA/T^2 \le 6\lambda^2 BMA/T$. The normalization is an exact algebraic identity and preserves the Lipschitz constant. The diagonal bound follows from $\|\phi(Z_i)\|_\infty \le \lambda B + \lambda\sqrt{M}\delta_i$. □

A smooth global cutoff. Fix once and for all a smooth nonincreasing function $\chi : \mathbb{R} \to [0,1]$ equal to one on $(-\infty, 2]$ and zero on $[5, \infty)$. For $T \ge 1$, set

$$a = \frac{\log(2m)}{T^2}, \quad s(z) = a^{-1}\log\sum_{j=1}^m e^{az_j^2}, \quad r_T(z) = \chi(s(z)/T^2).$$

Since $\|z\|_\infty^2 \le s(z) \le \|z\|_\infty^2 + T^2$, we have $r_T = 1$ on $\|z\|_\infty \le T$ and $r_T = 0$ on $\|z\|_\infty \ge \sqrt{5}T$. Direct differentiation gives absolute constants depending only on χ such that

$$\sup_z \sum_j |\partial_j r_T(z)| \le C/T, \qquad \sup_z \sum_{j,k} |\partial_{jk} r_T(z)| \le C\log(2m)/T^2.$$

Indeed, on the support of the derivatives of χ, every $|z_j|$ is at most $\sqrt{5}T$. If $p_j = e^{az_j^2}/\sum_k e^{az_k^2}$, then $\partial_j s = 2p_j z_j$ and $\partial_{jk} s = 2p_j \mathbf{1}_{j=k} + 4a(p_j z_j^2 \mathbf{1}_{j=k} - p_j p_k z_j z_k)$. Their entrywise derivative sums are at most $2\sqrt{5}T$ and $2 + 40\log(2m)$, respectively. The chain rule gives the assertion.

For any globally λ-Lipschitz activation ϕ with $\phi(0) = 0$, choose a fixed smooth nonnegative probability density ρ supported on $[-1,1]$ and mollify $\psi_\tau(t) = \int \phi(t - \tau y)\rho(y)\,dy$, where $0 < \tau \le 1$. Then $\|\psi_\tau - \phi\|_\infty \le \lambda\tau$, $|\psi_\tau'| \le \lambda$, and $|\psi_\tau''| \le \lambda\|\rho'\|_1/\tau$. For ReLU one may alternatively use $(t + \sqrt{t^2 + \tau^2})/2$, with uniform error at most $\tau/2$. For GELU one may use $\psi = \phi$ directly, since its first two derivatives are bounded by absolute constants. Write $b_x(z) = r_T(z)\psi(z_x)$. These bounded features have uniform derivative bounds

$$\begin{aligned}
B_0 &:= \max_x \|b_x\|_\infty \le C\lambda T, \\
B_1 &:= \max_x \sup_z \sum_j |\partial_j b_x(z)| \le C\lambda, \\
B_2 &:= \max_x \sup_z \sum_{j,k} |\partial_{jk} b_x(z)| \le C\lambda(\log(2m)/T + \tau^{-1}).
\end{aligned}$$

where for GELU one can set $\tau = 1$ in the last bound. In general, the empirical covariance of $b(Z_i)$ differs from Q_{bulk} by at most $C\lambda^2 T\tau$ in maximum norm, deterministically.

Lemma 6.3 (A sparse exponential-tilt certificate)**.** *Let $F = (F_1, \dots, F_D)$ be bounded observables of a probability law μ, with $|F_j| \le F_0$, and let $\widehat{F}$ be their empirical means on N independent draws. For $G, K > 0$ and a positive integer s, except on an event of probability at most*

$$\exp\{s \log(2D+1) - N(K - G^2 F_0^2/(2s))\},$$

there is a tilt $d\mu_\Gamma = e^{\Gamma \cdot F - \Lambda(\Gamma)} d\mu$ with $\|\Gamma\|_1 \le G$, relative entropy at most K, and $\|\widehat{F} - \mathbb{E}_{\mu_\Gamma} F\|_\infty \le K/G$. Here $\Lambda(\Gamma) = \log \mathbb{E}_\mu e^{\Gamma \cdot F}$.

Proof. Write $g(\Gamma) = \Gamma \cdot \widehat{F} - \Lambda(\Gamma)$. For every fixed Γ, independence gives $\mathbb{E} e^{N g(\Gamma)} = 1$. Every vector in the radius-G ℓ^1 ball is the mean of a random vertex from $\{0, \pm G e_1, \dots, \pm G e_D\}$. Average s independent such vertices to obtain Γ_s with $\mathbb{E}\Gamma_s = \Gamma$. For every fixed observation, Hoeffding's exponential lemma gives

$$\mathbb{E}_{\Gamma_s} e^{(\Gamma_s - \Gamma) \cdot F} \le e^{G^2 F_0^2/(2s)}.$$

Jensen's inequality and Fubini therefore show $\mathbb{E}_{\Gamma_s} \Lambda(\Gamma_s) \le \Lambda(\Gamma) + G^2 F_0^2/(2s)$. For at least one of the at most $(2D+1)^s$ sparse averages, $g(\Gamma_s) \ge g(\Gamma) - G^2 F_0^2/(2s)$. Exponential Markov and a union bound prove that $\max_{\|\Gamma\|_1 \le G} g(\Gamma) \le K$ outside the stated event.

Choose a maximizer Γ_* on the compact ball. The first-order optimality condition says $\Gamma_* \cdot \nabla g(\Gamma_*) = G \|\nabla g(\Gamma_*)\|_\infty$, including an interior maximum, where both sides are zero. Concavity and $g(0) = 0$ give $g(\Gamma_*) \ge \Gamma_* \cdot \nabla g(\Gamma_*)$. Since $\nabla g = \widehat{F} - \mathbb{E}_{\mu_\Gamma} F$, this proves the required approximation. Finally the tilt entropy is

$$D(\mu_{\Gamma_*} \| \mu) = g(\Gamma_*) - \Gamma_* \cdot \nabla g(\Gamma_*) \le K.$$

□

Lemma 6.4 (Polynomial stability of the tilted bounded bulk)**.** *Let b_x be any bounded C^2 features satisfying the preceding B_0, B_1, B_2 bounds. For a matrix Γ with $\|\Gamma\|_{1,\mathrm{ent}} \le G$, define*

$$\mathcal{F}_\Gamma(Q) = \frac{\mathbb{E}_{Z \sim \mathcal{N}(0,Q)}[b(Z) b(Z)^\top e^{b(Z)^\top \Gamma b(Z)}]}{\mathbb{E}_{Z \sim \mathcal{N}(0,Q)} e^{b(Z)^\top \Gamma b(Z)}}.$$

Put $F_0 = B_0^2$, $F_1 = 2B_0 B_1$, and $F_2 = 2B_1^2 + 2B_0 B_2$. For all positive semidefinite Q, Q',

$$\|\mathcal{F}_\Gamma(Q) - \mathcal{F}_\Gamma(Q')\|_{\max} \le C_G \|Q - Q'\|_{\max}, \quad C_G = \tfrac{1}{2}\{F_2 + 2GF_1^2 + 2F_0(GF_2 + G^2 F_1^2)\}.$$

Moreover, with s any positive integer, except on an event of probability

$$\exp\{s \log(2m^2+1) - N(K - G^2 F_0^2/(2s))\},$$

the empirical covariance of $b(Z_1), \dots, b(Z_N)$ is within K/G in maximum norm of some $\mathcal{F}_\Gamma(Q)$ with $\|\Gamma\|_{1,\mathrm{ent}} \le G$ and tilt entropy at most K.

Proof. Apply Lemma 6.3 to the $D = m^2$ bounded observables $F_{xy}(z) = b_x(z) b_y(z)$. This gives the approximation and exceptional probability, including the entropy bound.

For stability, interpolate $Q_t = (1-t)Q + tQ'$, set $V = b^\top \Gamma b$, and let $\mathbb{E}_t$ denote Gaussian expectation tilted by e^V. Gaussian differentiation and the quotient rule give

$$\frac{d}{dt}\mathbb{E}_t F = \frac{1}{2}\sum_{j,k} (Q' - Q)_{jk} \{\mathbb{E}_t[\partial_{jk} F + \partial_j F \partial_k V + \partial_k F \partial_j V] + \mathrm{Cov}_t(F, \partial_{jk} V + \partial_j V \partial_k V)\}.$$

For $F = b_x b_y$, its sup norm and first/second derivative entry sums are at most F_0, F_1, F_2. The corresponding derivative sums for V are at most GF_1, GF_2. Taking absolute values and summing proves the claimed bound. All functions and derivatives are bounded; adding ϵI and passing to the limit proves the statement for singular covariances as well. No reciprocal partition-function estimate is used, so the stability constant is polynomial, not exponential, in G. □

6.1 Stable finite realization of tail atoms

The next certificate avoids any lower bound on the nonzero eigenvalues of the covariance. It is used for both tail atoms and the final readout.

Lemma 6.5 (An ℓ^1 certificate for an RKHS vector)**.** *If $u \in \mathcal{H}_Q$, $\|u\|_Q \le 1$, and $\eta > 0$, there is a vector $\beta \in \mathbb{R}^{\mathcal{X}}$ such that*

$$\|Q\beta - u\|_\infty \le \eta, \qquad \|Q\beta\|_Q \le 1, \qquad \|\beta\|_1 \le \frac{1}{4\eta}.$$

Proof. In the finite-dimensional Hilbert space $\mathcal{H}_Q$, minimize $\frac{1}{2}\|v\|_Q^2$ subject to $\|v - u\|_\infty \le \eta$. A minimum exists, and u is a strictly feasible point. The multiplier conditions for the $2|\mathcal{X}|$ linear inequalities therefore apply. With the signs chosen so that $v = Q\beta$, complementary slackness says

$$\beta \cdot (u - v) = \eta\|\beta\|_1.$$

The left side is $\langle v, u\rangle_Q - \|v\|_Q^2$, and hence is at most $\|v\|_Q - \|v\|_Q^2 \le 1/4$. The other two claims follow from feasibility and the comparison with the feasible point u. The argument also covers singular Q, since the optimization takes place in $\mathcal{H}_Q$ itself. □

Proposition 6.6 (Stable realization with auxiliary activations)**.** *Let Q have diagonal entries at most $M \ge 1$, let $A \ge 1$ and $0 < \varepsilon \le 1$, and suppose*

$$R = \sum_{j=1}^{r} c_j \psi_j(v_j)\psi_j(v_j)^\top, \quad c_j \ge 0, \quad \sum_j c_j \le A, \quad \|v_j\|_Q \le 1, \quad \psi_j(0) = 0, \quad \mathrm{Lip}(\psi_j) \le \lambda.$$

Take

$$\eta = \frac{\varepsilon}{64A\lambda^2(\sqrt{M}+1)}, \qquad n = \left\lceil 128A^2\lambda^4 M^2 \varepsilon^{-2} \log(4m^2) \right\rceil.$$

Then R admits an approximation $R'(Q')$ within ε in maximum norm, uniformly over all positive semidefinite Q' satisfying $\|Q' - Q\|_{\max} \le \eta^2$, by n auxiliary neurons of the stated activation class. The covariance coefficients are A/n, and the incoming weights have the explicit bounds established below.

Proof. Sample pairs (v_j, ψ_j) with probabilities c_j/A, adding a zero pair for the remaining probability. The absolute entries of every sampled outer product are at most $\lambda^2 M$, so Hoeffding's inequality gives samples whose average times A approximates R within $\varepsilon/4$ entrywise. Apply the RKHS certificate to the sampled v_j, obtaining $\|\beta_j\|_1 \le 1/(4\eta)$ and $\|Q\beta_j - v_j\|_\infty \le \eta$. For every Q' with $\|Q' - Q\|_{\max} \le \eta^2$,

$$\|Q'\beta_j - v_j\|_\infty \le 5\eta/4, \qquad \|\psi_j(Q'\beta_j) - \psi_j(v_j)\|_\infty \le 5\lambda\eta/4.$$

It follows that

$$R'(Q') = \frac{A}{n}\sum_j \psi_j(Q'\beta_j)\psi_j(Q'\beta_j)^\top \quad \text{satisfies} \quad \|R'(Q') - R\|_{\max} \le \varepsilon.$$

The displayed constants leave ample room for the outer-product error.

If $Q' = \sum_\ell a_\ell h_\ell h_\ell^\top$ is a finite weighted feature realization, each new neuron has activation ψ_j and weights

$$W_{j\ell} = a_\ell \sum_y \beta_j(y) h_\ell(y), \qquad |W_{j\ell}| \le a_\ell \|h_\ell\|_\infty/(4\eta).$$

Its covariance coefficient is A/n. □

These are ordinary finite feedforward computations with neuron-specific activations; the common capacity theorem controls them uniformly without coding the activations. All intermediate values and coefficients obey polynomial bounds whenever the preceding realization does.

7 The deep mean-field endpoint

Throughout this section $\phi(0) = 0$ and ϕ is λ-Lipschitz, where $\lambda \ge 1$ is fixed. The original Bayesian model uses this same activation at every neuron. The finite reconstructions used only for counting may have different zero-at-zero λ-Lipschitz activations at different neurons; the preceding capacity theorem covers precisely this enlarged class. No claim that those auxiliary activations are additional parameters of the Bayesian learner is made.

At $\gamma = 1$, let $H = L - 1$, $m = 2^d$, and write

$$Q_0(x, y) = d^{-1} \bar{x} \cdot \bar{y}, \qquad Q_s(x, y) = N^{-1} \sum_{i=1}^{N} h_i^s(x) h_i^s(y).$$

Conditionally on the preceding layers, the N preactivation fields at layer s are independent Gaussians with covariance Q_{s-1}. This is exact at every finite width.

7.1 One-step reconstruction and its polynomial modulus

We record explicitly why the local constructions can be iterated without an exponential dependence on m. Fix $K \ge 1$, put

$$q = 4K, \qquad A = 4(q + \log 2),$$

and suppose the input covariance has diagonal at most $M \ge 1$. For a desired output accuracy $0 < \alpha \le 1$, set

$$B = \sqrt{2M \log(4m)}, \qquad T = \max\{1, 2B, 96\lambda^2 BMA/\alpha\}.$$

Use the smooth cutoff from the preceding section. Mollify ϕ at scale $\tau \le 1$ small enough that the maximum-norm difference between its cutoff covariance and that using the actual activation is at most $\alpha/16$, uniformly in the Gaussian covariance. For example $\tau = \min\{1, \alpha/[64C\lambda^2(T+1)]\}$ works with the absolute constant C in the smoothing estimate. Let F_0, F_1, F_2 be the bounded observable and derivative budgets in that section, and put

$$G = 16q/\alpha, \qquad s = \max\{1, \lceil G^2 F_0^2 \rceil\},$$

$$C_G = \tfrac{1}{2}\{F_2 + 2GF_1^2 + 2F_0(GF_2 + G^2F_1^2)\}, \qquad \eta = \frac{\alpha}{512A\lambda^2(\sqrt{M}+1)},$$

$$r(\alpha) = \min\{1, \eta^2, \alpha/[16(1 + C_G)]\}, \qquad N_0(\alpha) = \left\lceil \frac{2s\log(2m^2+1)}{q} \right\rceil.$$

Every displayed quantity is bounded by a polynomial in $K, M, \log(2m), \alpha^{-1}$, or, for r, its reciprocal is so bounded. The constants may depend on λ and the fixed smooth cutoff.

For $N \geq N_0(\alpha)$, except with probability at most $2e^{-NK}$, the empirical outgoing covariance is within $3\alpha/16$ of

$$\mathcal{F}_\Gamma(Q) + R, \qquad \|\Gamma\|_{1,\mathrm{ent}} \leq G,$$

where R is a mixture of unit-RKHS auxiliary-activation atoms with total coefficient mass at most A. Indeed, the tail decomposition error is at most $\alpha/16$, smoothing costs at most $\alpha/16$, and the entropy certificate costs at most $q/G = \alpha/16$. The projection failure is at most e^{-Nq}; the entropy failure is at most

$$\exp\{s\log(2m^2+1) - N(q-1/2)\} \leq e^{-NK}.$$

These bounds hold conditionally and uniformly in the input covariance.

Now suppose Q' is a positive semidefinite covariance with $\|Q' - Q\|_{\max} \leq r(\alpha)$, represented by finitely many weighted features from the preceding layer. The selected Γ and atom certificates remain fixed while Q is replaced by Q'. The bulk change is at most $\alpha/16$. The generic atom construction, with target accuracy $\alpha/8$, uses

$$n_A = \left\lceil 8192 A^2 \lambda^4 M^2 \alpha^{-2} \log(4m^2) \right\rceil$$

auxiliary neurons and approximates R within $\alpha/8$ at covariance Q'. Its certificate parameter is exactly the η above.

To realize the bulk, sample independent Gaussian rows of covariance Q' tilted by the bounded smooth observable. Evaluate the actual activation ϕ at those rows and absorb the scalar cutoff into their covariance coefficients. Each retained cutoff feature is bounded by $B_b = \lambda\sqrt{5}T$. Take

$$n_B = \left\lceil 1024 B_b^4 \alpha^{-2} \log(16m^2) \right\rceil.$$

Hoeffding's inequality and a union bound over m^2 entries give a realization with empirical covariance within $\alpha/16$ of its own expectation. The expectation differs from $\mathcal{F}_\Gamma(Q')$ by at most $\alpha/16$ because of the smooth activation approximation. Thus all the accumulated errors are less than α. This construction uses $n_A + n_B$ neurons, a polynomial number in $K, M, \log(2m), \alpha^{-1}$.

Finite weights and genuine feedforward evaluation. Write the preceding covariance as $Q' = \sum_{\ell=1}^{r} a_\ell h_\ell h_\ell^\top$, with $a_\ell \leq V := \max\{1, A\}$ and $\|h_\ell\|_\infty \leq B_h$. A bulk preactivation is realized with weights $\sqrt{a_\ell}\xi_\ell$, where the Gaussian vector ξ is tilted by a density ratio at most e^{2GF_0}. Intersecting the empirical covariance concentration event with a Gaussian-coordinate tail event ensures

$$|\xi_\ell| \leq R_\xi := \sqrt{4GF_0 + 2\log(16 n_B \max\{r, 1\})}$$

for every sampled coordinate. The latter event fails with probability at most $1/8$, whereas the preceding covariance event fails with probability less than $1/8$, so the intersection has positive probability. Bulk weights are therefore bounded by $\sqrt{V}R_\xi$. The atom weights are $a_\ell \sum_y \beta_j(y) h_\ell(y)$ and are bounded by $VB_h/(4\eta)$. All new covariance coefficients are at most V: they are $r_T(z)^2/n_B$ for bulk rows and A/n_A for atom rows.

For example, if B_W denotes the maximum of the two incoming-weight bounds, the actual hidden coordinates satisfy the deterministic bound $B_h^{\mathrm{new}} \leq \lambda r B_W B_h$. This bounds the untruncated physical activations as well as the weighted features. Hence at any fixed number of layers both the weights and all physical hidden coordinates remain polynomially bounded. The cutoff scalars, tilt matrices, and RKHS certificates select finitely many real weights; they are not evaluated on the input when the resulting network predicts. At the first layer use input features $h_\ell(x) = \bar{x}_\ell$ and coefficients $a_\ell = 1/d$. In particular an atom row is the ordinary linear form with weights $d^{-1}\sum_y \beta_j(y)\bar{y}$.

7.2 A uniform functional exceptional-mass bound

Proposition 7.1 (Deep mean-field functional compression)**.** *For each fixed L and λ, there are polynomials $P_{L,\lambda}$ and $J_{L,\lambda}$ such that the following holds for every $d \geq 1$, $K \geq 1$, and $N \geq P_{L,\lambda}(d,K)$. Except on an event of raw Gaussian probability at most $(2H+1)e^{-NK}$, every globally strict-margin draw has the same Boolean sign function as an auxiliary depth-L network with normalized energy at most $J_{L,\lambda}(d,K)$. All its neuronwise activations vanish at zero and have Lipschitz constant at most λ.*

Proof. Define layerwise deterministic diagonal bounds

$$M_s = 2[8\lambda^2\{\log(4m) + A + 1\}]^s, \qquad 0 \leq s \leq H.$$

At layer s, apply the one-step construction with $M = M_{s-1}$ and its corresponding projection cutoff B_s. On its projection-good event, the next covariance has diagonal at most $2\lambda^2 B_s^2 + 2\lambda^2 M_{s-1} A \leq M_s$. Starting from $\max_x Q_0(x,x) \leq 2$, induction therefore validates the diagonal budget used at each layer. Every M_s is polynomial in d, K for fixed L, λ because $\log m = d \log 2$. There are at most two exceptional events per hidden layer, as accounted for in the one-step construction.

Set the desired final covariance error to $\zeta_H = 1/(128K)$. Working backwards, for $s = H, H-1, \ldots, 1$, use the one-step construction with $\alpha = \zeta_s$ and $M = M_{s-1}$, and define $\zeta_{s-1} = r_s(\zeta_s)$ using that layer's modulus. Each ζ_s^{-1} is polynomially bounded in d, K at fixed L: this is a fixed finite composition of the polynomial reciprocal moduli above. Taking the maximum of the finitely many thresholds $N_{0,s}(\zeta_s)$ defines a polynomial width threshold $P_{L,\lambda}(d,K)$ (increased to at least one if necessary). Starting from the exact Q_0, reconstruct the covariances forwards. The transported certificates are fixed for the original covariances; their estimates hold uniformly for every admissible preceding reconstruction. Thus no independence assumption is imposed on those certificates. Induction yields a weighted-feature covariance $\widehat{Q}_H$ with

$$\|\widehat{Q}_H - Q_H\|_{\max} \leq 1/(128K).$$

The construction above bounds every layer width, every physical weight, and every physical activation by polynomials in d, K at fixed depth.

For the actual Gaussian readout $v_i \sim N(0, N^{-2})$, put $E_v = N\|v\|_2^2 = N^{-1}\chi_N^2$. Markov's inequality at exponential parameter $1/4$ gives

$$\Pr\{E_v > 8K\} \leq \exp[-N(2K - \tfrac{1}{2}\log 2)] \leq e^{-NK}.$$

On its complement the original output z belongs to $\mathcal{H}_{Q_H}$ and $\|z\|_{Q_H}^2 \leq 8K$. The scaled form of Lemma 6.5, with approximation accuracy $1/8$, gives

$$\|Q_H\beta - z\|_\infty \leq 1/8, \qquad \|\beta\|_1 \leq 16K.$$

Replacing Q_H by $\widehat{Q}_H$ adds at most $1/8$ to this error. For $\widehat{Q}_H = \sum_i a_i h_i h_i^\top$, the readout $v_i' = 2a_i \sum_y \beta(y) h_i(y)$ therefore represents a function with the same signs as z and strict margin greater than $1/2$. Its weights are bounded by $32KVB_h$.

Pad the finitely many reconstructed layers to their maximum width r_* with zero neurons, which is harmless since every activation vanishes at zero. If B_* bounds all weights, the normalized energy is at most

$$[d(d+1) + (L-1)r_*^2]B_*^2.$$

Taking a polynomial upper bound on this expression defines $J_{L,\lambda}(d,K)$. A conditional union bound over the H layers and the readout proves the stated raw exceptional probability. The auxiliary capacity theorem applies uniformly to the possibly different activations; no finite coding of activation formulas is needed. □

8 One polynomial for every scaling at least as strong as mean field

The endpoint and spectral arguments together give a stronger uniform statement than treating each exponent separately. Parameterize the raw Gaussian prior by $0 < t \le 1$, with coordinate variances

$$\frac{t^2}{d} \quad \text{in the first layer,} \qquad \frac{t^2}{N} \quad \text{in internal layers,} \qquad \frac{t^2}{N^2} \quad \text{at the readout.}$$

The scaling exponent $\gamma \ge 1$ corresponds to $t = N^{(1-\gamma)/2}$. Allowing every $t \in (0, 1]$ also permits the scale to be chosen independently at each width. Write $\nu_{N,t} = \mu_{N,\sigma=t/\sqrt{N}}$ for this raw law.

Lemma 8.1 (Uniformity of the compression ingredients)**.** *Let $A_f = A$ be the constant in Proposition 5.1. At every $N \ge C$ and $0 < t \le 1$,*

$$\nu_{N,t}(U_{N,f}) \ge e^{-A_f N/(2t^2)}.$$

The mean-field functional compression proposition remains true with the same polynomials $P_{L,\lambda}, J_{L,\lambda}$ and raw exceptional bound $(2H+1)e^{-NK}$ whenever $N \ge P_{L,\lambda}(d, K)$, uniformly for every $t \in (0, 1]$. Also, whenever

$$t \le a_K/8, \qquad t^{-2} \ge 8(d+1),$$

the spectral functional compression at energy $(L+3)K$ holds at every width, with raw exceptional mass at most $Le^{-NK/(8t^2)}$.

Proof. The target-cylinder proof uses scalar Gaussian variance t^2 in its constrained coordinates and $t \le 1$ in its deterministic error bound; replacing N^γ by N/t^2 gives the first assertion directly.

In the endpoint argument the conditional preactivation covariance is now $t^2 Q_{s-1}$ and the first one is $t^2 Q_0$. Apply the same one-step construction to these covariances. Their diagonal budgets do not increase. If Q'_{s-1} approximates Q_{s-1}, then

$$\|t^2 Q'_{s-1} - t^2 Q_{s-1}\|_{\max} \le \|Q'_{s-1} - Q_{s-1}\|_{\max}.$$

The weighted feature coefficients for the next preactivation are multiplied by t^2. Thus Gaussian incoming weights are multiplied by t, atom incoming weights by t^2, and all existing upper bounds on weights and feature magnitudes remain valid. Initially the input-feature coefficients are t^2/d. The readout energy is $E_v = t^2 \chi^2_N / N$, whose upper tail is no larger than at $t = 1$. Hence the same polynomial thresholds, compressed-energy bound, and raw exceptional probability apply. Although the reconstructed networks may depend on t, the single auxiliary energy class containing them does not depend on it.

For spectral compression, the weighted first-layer energy and readout energy are $(t^2/N)\chi^2_{N(d+1)}$ and $(t^2/N)\chi^2_N$; the internal entries have variance t^2/N. The two stated inequalities are exactly the hypotheses used in the spectral proof. Its singular-value, RMS-activation, and readout estimates therefore give the displayed tail, without a lower bound on N. □

Choose the following budgets, all independent of learner width and scale:

$$\begin{aligned}
K_s &= 4A_f + 8\log(2L/\varepsilon), & t_* &= \min\{a_{K_s}/8, [8(d+1)]^{-1/2}\},\\
K_m &= 2 + A_f/t_*^2 + \log\{2(2H+1)/\varepsilon\}, & N_* &= \lceil \max\{1, P_{L,\lambda}(d, K_m)\} \rceil,\\
D_* &= d + 2 + (L-2)N_*, & J_0 &= 2A_f + 2D_* \log 2 + 4\log(2/\varepsilon),\\
J_* &= (L+3)K_s + J_{L,\lambda}(d, K_m) + J_0.
\end{aligned}$$

Each budget, including t_*^{-1}, is polynomially bounded in $d, C, \log(1/\varepsilon)$ for fixed L, λ.

Theorem 8.2 (All widths and all scales at least as strong as mean field)**.** *Fix a depth $L \geq 2$ and an activation ϕ with $\phi(0) = 0$ and Lipschitz constant at most λ. Suppose a Boolean target has a width-C, depth-L implementation with weights bounded by C and strict global margin $1/2$. Let the original fixed-ϕ Gaussian prior have the variances displayed above, condition it on an absolute margin $1/2$ at every Boolean input, and then condition on exact training labels. For every input distribution on the Boolean cube and every $0 < \varepsilon, \delta < 1$, one event of training-sample probability at least $1 - \delta$ gives posterior average sign error at most ε* simultaneously for every integer $N \geq C$ and every $0 < t \leq 1$, *provided*

$$n \geq \frac{2}{\varepsilon}\left[\mathcal{B}(J_*) + \log\frac{1}{\delta}\right].$$

In particular, one fixed-depth polynomial sample bound works for every $\gamma \geq 1$, including $\gamma = 1$, without a polynomial degree or constant that deteriorates as $\gamma \downarrow 1$. ReLU, exact GELU, SiLU, tanh, and every other fixed Lipschitz activation vanishing at zero are covered whenever the stipulated teacher exists. For the correct label to have posterior probability strictly greater than $2/3$ outside a test set of mass at most ε, apply the bound with average-error accuracy $\varepsilon/3$.

Proof. Let $\mathcal{F}_*$ be the single auxiliary strict-margin function class of energy at most J_*. We show that at every $N \geq C$ and $0 < t \leq 1$, the raw probability of margin-separated functions outside this class, divided by the raw target probability, is at most $\varepsilon/2$.

If $t \leq t_*$, spectral compression gives the ratio bound

$$L\exp\left[-\frac{N}{t^2}(K_s/8 - A_f/2)\right] \leq \varepsilon/2.$$

This applies at every width. If $t \geq t_*$ and $N \geq N_*$, the uniform endpoint compression gives

$$(2H+1)\exp\left[-N\left(K_m - \frac{A_f}{2t^2}\right)\right] \leq \varepsilon/2.$$

For the remaining bounded widths $C \leq N < N_*$, use total energy instead of a union over the continuum of possible scales. The parameter count is $p = N(d+2) + (L-2)N^2$, so $p/N \leq D_*$. The normalized energy has law $\kappa_N = (t^2/N)\chi^2_p$. Its exponential moment at $N/(4t^2)$ is $2^{p/2}$, and therefore

$$\begin{aligned}\frac{\nu_{N,t}\{\kappa_N > J_0\}}{\nu_{N,t}(U_{N,f})} &\leq \exp\left\{-\frac{N}{t^2}(J_0/4 - A_f/2) + \frac{p}{2}\log 2\right\} \\ &\leq \exp\left\{-\frac{N}{t^2}(J_0/4 - A_f/2 - D_*\log 2/2)\right\} \leq \varepsilon/2.\end{aligned}$$

Here $t \leq 1$ and $N \geq 1$ justify the last two inequalities. A globally margin-separated draw with $\kappa_N \leq J_0$ itself belongs to the enlarged auxiliary class. Thus all three regimes are contained in the same $\mathcal{F}_*$ apart from the asserted relative mass.

The target must belong to $\mathcal{F}_*$, since otherwise the relative mass would be at least one. Normalization by the global margin event cancels, and conditioning on any noiseless training labels preserves the upper bound $\varepsilon/2$ on posterior mass outside $\mathcal{F}_*$. The capacity theorem bounds $\log|\mathcal{F}_*|$ by $\mathcal{B}(J_*)$. A finite-class union bound ensures, on one sample event of probability at least $1 - \delta$, that every consistent member has true error at most $\varepsilon/2$. That event refers only to the single function class, so it works for the entire continuum of scales and every learner width simultaneously. Adding the posterior mass of the complement gives the claimed average-error bound. □

9 Exact GELU implementations of Boolean circuits

The preceding theorems apply to small implementing networks. To translate ordinary Boolean circuits into that premise, take a layered fan-in-two circuit with NOR, COPY, and NOT gates and one constant wire. Every layer has width at most C, counting that wire. Inputs are encoded as bits $b_i = (1 + x_i)/2$.

For ReLU, a NOR gate is $\mathrm{ReLU}(1 - a - b)$, a copy is $\mathrm{ReLU}(a)$, and NOT is $\mathrm{ReLU}(1 - a)$ on Boolean inputs. These operations preserve exact bits; a constant wire of value one supplies offsets.

Proposition 9.1 (Exact same-depth GELU compilation)**.** *A layered Boolean circuit with H stages and width C, counting its constant wire, has an exact GELU implementation with H hidden layers, hidden width at most $2C$, and individual weights bounded by 4. Its real-valued output can be chosen to equal the target sign ± 1 exactly. Consequently the learning theorems apply with teacher complexity at most $\max\{4, 2C\}$.*

Proof. Write $\tau = \Phi(1)$, $s = 2\tau - 1$, $A = \tau/s$, and $B = (1 - \tau)/s$. The elementary bound $\tau > 2/3$ gives $1 < A < 2$ and $0 < B < 1$. Direct substitution shows

$$A\,\mathrm{GELU}(u) + B\,\mathrm{GELU}(-u) = \mathrm{ReLU}(u) \qquad (u \in \{-1, 0, 1\}). \tag{4}$$

At $u = 1$, the left side is $A\tau - B(1 - \tau) = 1$; at $u = -1$ it is $-A(1 - \tau) + B\tau = 0$; and at zero it vanishes.

Represent each gate by two GELU neurons with preactivations u and $-u$. Decode the gate's bit as the linear combination with coefficients A, B. For NOR take $u = 1 - a - b$, for COPY $u = a$, and for NOT $u = 1 - a$. These always lie in $\{-1, 0, 1\}$. Substitute the preceding layer's linear decoders into each new preactivation; this implements all stages without an extra hidden layer. A single constant neuron has value τ and is propagated by incoming weight $1/\tau$, so it supplies the required constant one in each preactivation. The first stage uses the affine bit encoding above.

Two copies of the same gate input produce coefficients of size at most $2A < 4$; all other internal coefficients have magnitude at most 2 or $1/\tau < 3/2$. The final linear readout $2b - 1$ has coefficients $2A, 2B, -1/\tau$, all bounded by 4. The total width is at most twice the original width, and the output is exactly ± 1. Pad to width $\max\{4, 2C\}$ when a teacher complexity convention also requires weights bounded by the width. □

The identity in (4) is required only on the three gate preactivations, not on the whole real line. Thus this is an exact discrete compilation using fixed coefficients, with no large-gain approximation or homogeneity assumption. The derivative $\Phi(t) + t(2\pi)^{-1/2}e^{-t^2/2}$ has absolute value less than 2, verifying the activation hypothesis directly.

10 Scope and remaining questions

The result separates two obstacles caused by unbounded activations. Large individual neuron values are handled by Euclidean norm control in the capacity argument. At intermediate exponents $1 < \gamma < 2$, large total Gaussian parameter energy is handled by retaining the part with large singular values; discarded matrices have small operator norm and their accumulated output error can be bounded at fixed depth.

At the deep mean-field endpoint, the Gaussian bulk has operator norm of constant order. The proof therefore keeps its covariance contribution, using a bounded tilt and separate tail atoms. The covariance reconstruction is polynomial at each fixed depth but repeatedly composes polynomial moduli. It does not give a polynomial-depth guarantee or a competitive numerical bound.

The surrogate networks used for counting may have different zero-at-zero Lipschitz activations at different neurons. This is a proof device, not a change to the Bayesian model. In particular, the argument does not establish a same-activation circuit-compression converse for the generic activation class.

The global-margin conditioning is also substantive. It converts the real-output capacity bound into a finite class of Boolean functions and lets us compare raw target mass with raw exceptional mass. Removing it, changing the likelihood, or learning real outputs requires additional arguments.

Provenance and verification. This paper extends the collaborative strong-prior setup and Gaussian-prior learning program, including contributions from Claude, to unbounded activations. The capacity, spectral, mean-field reconstruction, and exact GELU compilation arguments have undergone independent AI-agent mathematical review. The main learning statements have Lean proofs; the repository records their exact scope and exclusions.

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